

Team 3 GDD

1. Core Loop in One Sentence

Author: Kai, Kayla, Samaii, Daryl

Players navigate a pitch-black childhood home using limited acoustic pulses to complete domestic tasks that unlock the front exit, managing their noise and breath to escape before a blind, sound-hunting monster tracks them down.

2. The Systems (One Per Member)

Author: Kai, Kayla, Samaii, Daryl

System 1: Acoustic Pulse & Perception System

- **Owner:** Kai
- **State It Owns:**
 - `pulse_charges` (int: 0 to 5)
 - `is_pulse_active` (bool)
 - `pulse_origin` (Vector3: world coordinates)
 - `pulse_radius` (float: current expansion distance)
 - `active_noise_events` (Array of Dictionary/Struct: {`position`: Vector3, `intensity`: float, `timestamp`: float})
- **What It Does:** Manages player-triggered echolocation pulses, visual wave expansion, pulse charge inventory, and the global dispatch of noise events into the game level whenever the player pulses, steps, or interacts.

System 2: Monster AI & Behavior System

- **Owner:** Daryl
- **State It Owns:**
 - `monster_state` (enum: DORMANT, INVESTIGATING, HUNTING, STALKING)
 - `investigation_target` (Vector3: world coordinates)
 - `current_speed` (float)
 - `aggro_level` (float: 0.0 to 1.0)
 - `kill_proximity_active` (bool)
- **What It Does:** Governs monster pathfinding navigation across the house navmesh, evaluates sensory stimulus priorities, transitions behavioral hunting states, and executes the game-over kill strike when catching the player.

System 3: Familiar House & Environment System

- **Owner:** Samaii
- **State It Owns:**
 - `current_surface_type` (enum: CARPET, HARDWOOD, TILE, CLUTTER)
 - `resonance_multiplier` (float: 0.2 to 2.0)
 - `room_layout_state` (Dictionary: door locks, open/closed doorways, blocked hazard zones)

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- **objective_progress** (Dictionary: {key_found: bool, bolt_unlocked: bool, chain_cleared: bool})
- **What It Does:** Tracks player footing colliders, assigns surface sound dampening or amplification, manages physical room doors, squeaky floorboards, and tracks progress toward unlocking the exit.

System 4: Sanity & Breath Control System

- **Owner:** Kayla
- **State It Owns:**
 - **lung_capacity** (float: 0.0 to 100.0)
 - **is_holding_breath** (bool)
 - **panic_level** (float: 0.0 to 1.0)
 - **heart_rate_bpm** (int: 60 to 180)
 - **is_gasps_active** (bool)
- **What It Does:** Tracks breath stamina depletion and replenishment, increases panic when the monster is near, triggers an uncontrollable auditory gasp when lung capacity hits zero, and feeds audio/visual heartbeat distortion to the player camera.

3. The Seams

Author: Kai, Kayla, Samaii, Daryl

Seam 1: Acoustic Perception → Monster AI

- **Writer:** Acoustic Pulse & Perception System (Kai) writes **pulse_origin** and **active_noise_events**.
- **Reader:** Monster AI & Behavior System (Daryl).
- **Reader Decision:** When a new **NoiseEvent** is published, the Monster AI compares event intensity against its hearing threshold. If audible, it aborts its current patrol, sets its **investigation_target** directly to **pulse_origin**, and transitions to **INVESTIGATING**.
- **Freeze Test:** If **active_noise_events** is frozen to empty, the monster never reacts to player pulses and remains stuck in its default patrol path.

Seam 2: Environment System → Acoustic Perception System

- **Writer:** Familiar House & Environment System (Samaii) writes **resonance_multiplier**.
- **Reader:** Acoustic Pulse & Perception System (Kai).
- **Reader Decision:** When an echo pulse or movement step occurs, the perception system multiplies the baseline noise radius and intensity by **resonance_multiplier** before publishing the resulting **NoiseEvent** (e.g., doubling the detection radius on bare hardwood or halving it on carpet).
- **Freeze Test:** If **resonance_multiplier** is frozen at 1.0, carpet runners lose their stealth utility, and stepping on hardwood no longer triggers a sudden monster sprint.

Seam 3: Breath Control → Acoustic Perception System

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- **Writer:** Sanity & Breath Control System (Kayla) writes `is_gasps_active`.
- **Reader:** Acoustic Pulse & Perception System (Kai).
- **Reader Decision:** When `is_gasps_active` transitions to `true`, the acoustic system instantly instantiates a high-priority, maximum-intensity `NoiseEvent` at the player's exact world coordinates, bypassing player input.
- **Freeze Test:** If `is_gasps_active` is frozen to `false`, players can stay hidden indefinitely next to the monster without ever being forced to breathe or give away their location.

Seam 4: Monster AI → Sanity & Breath Control System

- **Writer:** Monster AI & Behavior System (Daryl) writes `monster_world_position` and `monster_state`.
- **Reader:** Sanity & Breath Control System (Kayla).
- **Reader Decision:** Measures the distance to the monster. If the monster is within 4 meters, it accelerates `lung_capacity` drain by 2x and drives `panic_level` toward 1.0.
- **Freeze Test:** If monster position is frozen far away, breath-holding stamina drains at a constant, relaxed rate regardless of whether the monster is in the room.

4. The Familiar Thing, The Turn, and Player Action

Author: Kai, Kayla, Samaii, Daryl

- **The Familiar Thing:** The childhood family home at night. The player relies on instinctual spatial memory: the exact number of paces between bedrooms, the location of the stairs, the runner rug in the hallway, and which doors stay open.
- **The Turn:** The safe, trusted layout becomes an acoustic death trap. Total darkness conceals altered physical conditions: familiar doors are deadbolted from the inside, floorboards creak underfoot, and the only tool that allows the player to see—an acoustic pulse—acts as a homing flare for a lethal entity.
- **How That Changes What the Player Does (Load-Bearing):** Players cannot rely on passive eyesight or fast sprinting. They must discard their mental map, treat open hardwood areas as dangerous chasms, route their path exclusively across dampening rugs, and use echo pulses as sacrificial distraction flares to lure the monster away from escape objectives.

5. The Monster

Author: Kai, Kayla, Samaii, Daryl

Monster Rules

- **Rule 1: The Ghost Coordinate (Origin Anchor)**
 - **System State Read:** Reads `pulse_origin` (Vector3) from the **Acoustic Pulse & Perception System** (Kai).
 - **The Rule:** The monster does not pathfind toward the player's moving avatar; it paths exclusively toward the exact (x, y) world coordinate where a noise event

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originated, investigating that specific anchor point for 2.5 seconds before returning to patrol.

- **How Players Figure It Out:** Through trial and dying. A player will inevitably panic, tap an echo pulse in a doorway, and dash around a corner onto a carpeted surface. Instead of rounding the corner to kill them, the monster will barrel directly into the empty doorframe where the pulse occurred, swipe at the empty space, and sniff around. Players realize that an echo reveals their *past* location, not their current one—turning the pulse into a deliberate distraction flare.
- **Rule 2: Resonance Amplification (Surface Echo Penalty)**
 - **System State Read:** Reads `current_surface_resonance` (a float multiplier from 0.2 to 2.0) from the **Familiar House & Environment System** (Samaii).
 - **The Rule:** Hard floor surfaces (hardwood, bathroom tile, loose clutter) create high-amplitude vibrations that instantly send the monster into a sprint, while dampening surfaces (runner rugs, bath mats, carpet) halve its detection radius and keep it in a slow stalking gait.
 - **How Players Figure It Out:** Audio cues and behavioral contrast. When clapping or stepping on bare hardwood, the monster lets out an immediate shriek and closes the distance in seconds. When doing the exact same action while standing on the living room area rug, the monster merely grunts and wanders slowly toward the hallway. Observant players notice that rugs act as safe acoustic islands where they can pulse with much less risk.
- **Rule 3: The Gasp Response (Vocal Freeze)**
 - **System State Read:** Reads `is_holding_breath` (boolean) and `vocal_noise_level` (float) from the **Sanity / Breath Control System** (Kayla).
 - **The Rule:** When the monster comes within proximity (under 2 meters), it cannot detect static collision or player touch, but it enters an acute sensory lock: any audio pulse or involuntary breathing vocalization triggers an instant, un-dodgeable kill strike. Holding breath guarantees survival even if the monster physically brushes past the player.
 - **How Players Figure It Out:** Players trapped in a dead-end room will naturally stop moving. If they don't hold their breath, their character's stamina meter depletes, triggering a forced audible gasp—followed immediately by an instant kill. In subsequent runs, players who hold their breath notice the monster walking right along their collision box without triggering an attack, learning that proximity alone isn't lethal—only breathing is.

Scariest Moment

- **Location:** Upstairs Hallway (outside the linen closet)
- **Rules in Play:** Rule 1 (Ghost Coordinate), Rule 2 (Resonance Amplification), Rule 3 (Gasp Response)

You run out of steps in the dark and fire an echo to find the stairs. The pulse illuminates the hallway—and reveals the monster standing inches from your face.

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You scramble back toward the linen closet, but your foot hits bare hardwood, triggering a loud resonance spike (Rule 2) while the monster charges the doorway you just pulsed from (Rule 1). You dive inside the closet and hold your breath. The screen fades to black, and the monster stops right in the open doorway, inches away, sniffing the air—leaving you pinned and praying your lung meter doesn't run out before it turns around (Rule 3).

6. Art Direction Sheet

Author: Kai, Kayla, Samaii, Daryl

- **Visual Style:** Minimalist acoustic wireframe / lidar aesthetic. The default screen is pitch black. Pulses generate expanding, fading monochrome edge lines outlining rooms and props.
- **Color Palette:**
 - Background Void: Deep Pitch Black (#0a0a0c)
 - Acoustic Echo Lines: Spectral Cold White (#eef2f7)
 - Surface Dampening (Carpet): Slate Charcoal (#3d4852)
 - High-Resonance Hazard (Hardwood/Tile): Warning Ochre (#c98a2c)
 - Critical Panic / Death: Deep Crimson (#5c1414)
- **Lighting:** Zero global ambient light; zero standard point lights. Visuals are driven by a custom spatial pulse shader calculating distance from the active [pulse_origin](#).
- **Audio Direction:** Diegetic, binaural soundscape. Muffled heartbeats, floorboard creaks, panting breath, and sharp metallic clicking from the monster. No synthetic ambient music.
- **Camera Perspective:** First-person, 75° field of view, slight head bob, with a darkening tunnel-vision vignette tied to lung capacity depletion.

7. Scope & Three Cuts

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What Is In

- One complete, two-story house interior (Foyer, Living Room, Kitchen, Upstairs Hallway, Master Bedroom).
- 4 discrete member systems with 4 functional, freeze-tested seams.
- 3 surface types (Carpet, Hardwood, Creaky Clutter).
- 3 unspoken monster rules with distinct audio/visual telegraphs.
- 2-step escape objective loop: Find the hidden brass key downstairs, disarm the front door deadbolt, and exit.

Three Cuts

1. **Cut 1: Throwable Distraction Items:** Scrapped physics-based clutter items (mugs, books) to keep all sound events tied strictly to player locomotion and pulse resource management.

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2. **Cut 2: Multiple Monster Types:** Scrapped secondary roaming entities (like ghost family members) to focus entirely on perfecting the behavior tree and rule predictability of "The Listener."
3. **Cut 3: Dynamic Procedural Room Shifting:** Scrapped shifting walls in favor of a fixed, authentic home layout with scripted door barricades to maintain the authentic "familiar" feeling.